

AWS MACHINE LEARNING ENGINEER – ASSOCIATE

Amazon S3 Event Notifications and Performance Optimization

week 3

This presentation explains how S3 can automatically react to object events and how AWS improves performance for large files and high request workloads.

Professor: Victor Govindaswamy
Presented by: Rushikesh Nayak

In week 3, the main idea is to understand how S3 can become part of an automated data pipeline instead of acting only as a place to store files.

1. S3 Event Notifications

S3 can generate events when objects are created, removed, restored, or involved in replication. These events can automatically start another action.

2. Event Destinations

S3 events can be sent to AWS Lambda, Amazon SQS, or Amazon SNS. EventBridge can also provide more flexible routing and filtering.

3. S3 Performance

S3 automatically scales to high request rates. Performance can be increased further by distributing requests across multiple prefixes.

4. Large File Optimization

Multipart Upload, Transfer Acceleration, and Byte-Range Fetches help applications upload and download large objects more efficiently.

What I can explain while presenting:

I would introduce this week as two connected topics: automation and performance. In an ML project, new data may arrive in S3 automatically, and then AWS services can process that data without waiting for a person to start the next step.

Definition S3 Event Notifications allow Amazon S3 to send a notification when a selected event happens to an object inside a bucket.

- ObjectCreated – occurs when a new object is uploaded or created in the bucket.
- ObjectRemoved – occurs when an object is removed or deleted.
- ObjectRestore – occurs when an archived object is restored.
- Replication – can notify the application about replication-related events.
- The course notes that event notifications are usually delivered within seconds, but sometimes delivery can take a minute or longer.

Why is this useful?

Without events, an application may need to keep checking the bucket repeatedly. With event notifications, S3 can inform another AWS service as soon as an important change occurs.

Simple ML example

A training-image bucket receives a new image. S3 creates an ObjectCreated event, and a processing function can immediately validate or resize that image.

What I can explain while presenting:

I would explain an event notification like a doorbell. Instead of checking the door every few seconds, the bell tells us when somebody arrives. In the same way, S3 can tell another service when a new object arrives.

S3 does not always need to trigger the same action for every file. Object-name filtering can make the event more specific.

Example files in the bucket

photo01.jpg

report.txt

training.csv

photo02.jpg

Filter: *.jpg

Only matching image files
create the required notification.

Why filtering matters

Filtering reduces unnecessary processing. If a Lambda function is designed only for images, text or CSV files should not cause that function to run.

Course example

The course uses thumbnail generation as an example. When an image is uploaded, an S3 event can start the process that creates a smaller thumbnail.

What I can explain while presenting:

I would tell the audience that filtering helps us control which files start the workflow. This becomes important when one S3 bucket contains several types of data.

The course identifies three common direct destinations for S3 event notifications.

AWS Lambda

Lambda runs code in response to an event. This is useful when the next step requires immediate processing, such as resizing an image or validating an uploaded file.

Amazon SQS

SQS stores messages in a queue. This separates the arrival of new data from the speed of downstream processing, which is useful when many files arrive at once.

Amazon SNS

SNS publishes notifications to subscribers. It is useful when multiple systems or recipients need to receive information about the same S3 event.

Easy way to remember: Lambda = run code | SQS = queue work | SNS = publish a notification

What I can explain while presenting:

I would compare the three destinations instead of memorizing them separately. If I need code to run, I think Lambda. If I need to hold work safely, I think SQS. If I need to broadcast information, I think SNS.

Configuring an event is not enough. The destination also needs permission to accept the event from Amazon S3.

- Lambda uses a Lambda resource policy so that Amazon S3 is allowed to invoke the function.
- SNS uses an SNS resource/access policy so that S3 can publish a message to the topic.
- SQS uses an SQS resource/access policy so that S3 can send a message to the queue.
- If the event configuration looks correct but the destination never receives the event, permissions are an important place to check.

Important AWS point

These are resource-based permissions on the destination. They define whether S3 is trusted to perform the required action.

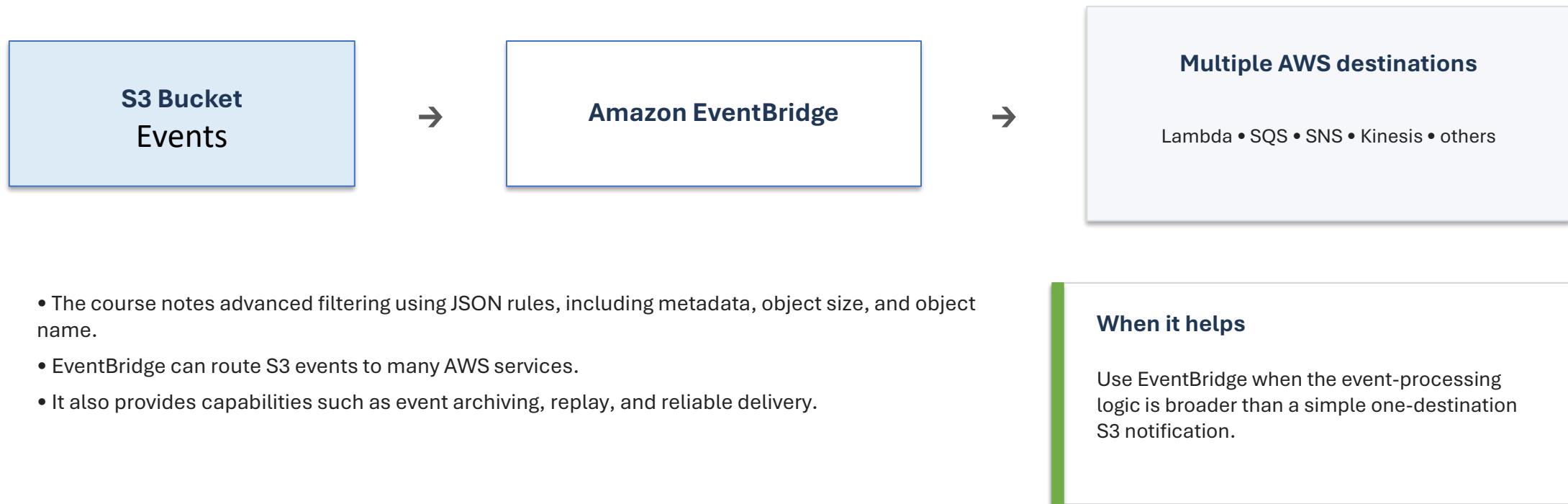
Exam-style thinking

A question may describe S3 events that are configured correctly but are not reaching Lambda, SNS, or SQS. The missing resource policy can be the cause.

What I can explain while presenting:

I would explain that AWS services still need authorization when they communicate with each other. The event tells S3 what to do, while the resource policy gives S3 permission to do it.

EventBridge gives S3 a more flexible event-routing option when the workflow needs advanced filtering or several possible destinations.



What I can explain while presenting:

I would describe EventBridge as a central event router. S3 produces the event, EventBridge examines it using rules, and then sends it to the correct destination.

Amazon S3 automatically scales to high request rates. The course lists latency around 100–200 milliseconds.

Write-type requests

At least 3,500 PUT, COPY, POST, or DELETE requests per second per prefix.

Read-type requests

At least 5,500 GET or HEAD requests per second per prefix.

What is a prefix?

A prefix is the beginning portion of an object key path. For example, in bucket/folder1/sub1/file, the course identifies /folder1/sub1/ as the prefix.

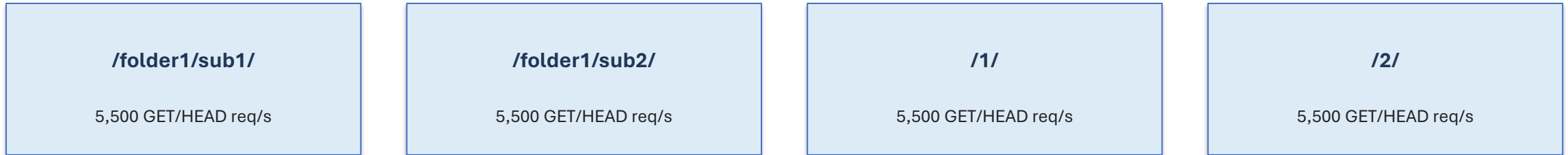
Important: there is no limit to the number of prefixes in a bucket, so total request throughput can scale across multiple prefixes.

What I can explain while presenting:

I would explain that the performance numbers are per prefix, not necessarily for the entire bucket. That is why spreading requests across several prefixes can increase total throughput.

Example: Scaling Reads Across Multiple Prefixes

The course provides an example using four different prefixes. Each prefix can support at least 5,500 GET/HEAD requests per second.



$4 \times 5,500 = 22,000$ GET/HEAD requests per second

What this means

If read traffic is spread evenly across all four prefixes, the example reaches 22,000 GET/HEAD requests per second in total.

Important understanding

The main lesson is that S3 can scale horizontally across prefixes. The exact object naming strategy should still match the application's needs.

What I can explain while presenting:

I would calculate the example on the slide so the audience can see where 22,000 comes from. Four prefixes multiplied by 5,500 read requests gives the total.

Definition Multipart Upload divides a large file into multiple smaller parts. Those parts can be uploaded independently and in parallel.



- The course recommends multipart upload for files larger than 100 MB.
- Multipart upload must be used for files larger than 5 GB.
- Uploading parts in parallel can make the overall transfer faster.
- Multipart Upload is also compatible with S3 Transfer Acceleration.

ML example

A very large training dataset or model artifact can be divided into parts and uploaded simultaneously instead of sending one huge file as a single transfer.

What I can explain while presenting:

I would explain multipart upload using a large package. Instead of moving one extremely heavy package at once, we divide it into smaller pieces and move several pieces at the same time.

S3 Transfer Acceleration is designed to improve transfer speed over long geographic distances.



- The file first travels to a nearby AWS edge location.
- AWS then forwards the data to the S3 bucket in the target Region.
- This can reduce the amount of the long-distance path that depends on the public internet.
- The course notes that Transfer Acceleration is compatible with Multipart Upload.

Example

A team in the United States needs to upload a very large ML dataset to an S3 bucket located far away, such as Australia. Transfer Acceleration may improve that long-distance transfer.

What I can explain while presenting:

I would emphasize distance. Multipart Upload solves the large-file problem, while Transfer Acceleration focuses on improving the network path for long-distance transfers.

Definition A byte-range request asks S3 for only a specific range of bytes from an object instead of retrieving the entire object in one request.

Part 1

Part 2

Part 3

Part 4

Part 5

Parallel GET requests

Different byte ranges can be requested in parallel. This can speed up downloading a large object.

Better resilience

If one range request fails, the application can retry that range instead of restarting the entire large download.

Partial data retrieval

The application can retrieve only the required part of the file, such as a header or the first section.

What I can explain while presenting:

I would explain that sometimes an application does not need the whole file. Byte-range fetches can save time by requesting only the useful section or by downloading several sections in parallel.

Scenario: an ML project receives new training images from users and stores the raw images in Amazon S3.

1. Upload

A new .jpg image is uploaded to the S3 bucket.

2. S3 Event

ObjectCreated is generated for the new image.

3. Lambda

A Lambda function can validate or resize the image.

4. SQS

A queue can hold additional processing work if many images arrive.

5. ML Data

Processed images can later be used by the training pipeline.

Why this is useful

What I can explain while presenting:

This example connects the services together. S3 stores the image, the event starts the workflow, Lambda can perform immediate processing, and SQS can help when downstream work needs to be buffered.

A useful exam approach is to identify the exact problem first and then select the S3 feature that directly addresses that problem.

A new object should automatically start processing	S3 Event Notification
A very large file must be uploaded	Multipart Upload
A file travels a long geographic distance	S3 Transfer Acceleration
A large download should be parallelized	Byte-Range Fetch
Only part of a file is required	Byte-Range Fetch
Very high S3 request throughput is needed	Distribute requests across prefixes

What I can explain while presenting:

I would read the requirement first. AWS exam questions often hide the answer inside words such as large file, long distance, partial download, automatic processing, or high request rate.

- S3 Event Notifications allow applications to react when important object events occur.
- Object-name filtering can limit notifications to relevant files such as .jpg images.
- Lambda, SQS, and SNS are common S3 event destinations, and each serves a different purpose.
- EventBridge adds advanced event filtering, routing, archiving, and replay capabilities.
- S3 performance scales per prefix, and multiple prefixes can increase aggregate throughput.
- Multipart Upload helps with large files, Transfer Acceleration helps with long-distance transfers, and Byte-Range Fetches help with parallel or partial downloads.

Main takeaway

Amazon S3 is more than storage. It can start automated workflows when data changes and provides several techniques for handling demanding data-transfer patterns in ML applications.

What to remember for the exam

Match the requirement to the service feature. Focus on why a feature is used rather than memorizing only its name.

What I can explain while presenting:

I would close by connecting both parts of the week: event notifications make S3 responsive, and performance features help S3 handle large files and high-volume workloads more efficiently.